6.3000 Quiz 2 — formulas to copy onto your handwritten sheets (6.3000 conventions: $\frac{1}{T}$, $\frac{1}{N}$ in analysis)

Four Fourier representations

	analysis	synthesis
CTFS	$a_k = \frac{1}{T}\int_T f(t)e^{-jk\omega_0 t}dt$	$f(t) = \sum_k a_k e^{jk\omega_0 t}$
DTFS	$a_k = \frac{1}{N}\sum_{n=\langle N\rangle} f[n]e^{-jk\Omega_0 n}$	$f[n] = \sum_{k=\langle N\rangle} a_k e^{jk\Omega_0 n}$
CTFT	$X(\omega) = \int x(t)e^{-j\omega t}dt$	$x(t) = \frac{1}{2\pi}\int X(\omega)e^{j\omega t}d\omega$
DTFT	$X(\Omega) = \sum_n x[n]e^{-j\Omega n}$	$x[n] = \frac{1}{2\pi}\int_{2\pi} X(\Omega)e^{j\Omega n}d\Omega$
DFT	$X[k] = \frac{1}{N}\sum_{n=0}^{N-1} x[n]e^{-j\frac{2\pi kn}{N}}$	$x[n] = \sum_{k=0}^{N-1} X[k]e^{j\frac{2\pi kn}{N}}$

$\omega_0 = \frac{2\pi}{T}$, $\Omega_0 = \frac{2\pi}{N}$. Trig form: $c_0 = a_0$, $c_k = 2\,\mathrm{Re}\,a_k$, $d_k = -2\,\mathrm{Im}\,a_k$.
DFT = $\frac{1}{N}\times$ DTFT samples: $X[k] = \frac{1}{N}X(2\pi k/N)$. $X[0]$ = average.

Properties (FT; same idea for FS, DTFT, DFT)

shift	$x(t-t_0) \to e^{-j\omega t_0}X$; FS: $a_k e^{-jk\omega_0 t_0}$
modulate	$e^{j\omega_c t}x \to X(\omega-\omega_c)$; DFT: $X[k-k_0]$
cos mod	$x\cos\omega_c t \to \frac{1}{2}X(\omega-\omega_c)+\frac{1}{2}X(\omega+\omega_c)$
scale	$x(at) \to \frac{1}{\lvert a\rvert}X(\omega/a)$
flip	$x(-t) \to X(-\omega)$; FS/DFT: a_{-k}
derivative	$x' \to j\omega X$; FS: $jk\omega_0 a_k$
times t	$t\,x(t) \to j\frac{dX}{d\omega}$
convolve	$x*h \to XH$; $xh \to \frac{1}{2\pi}X*H$
DFT product	$X[k]H[k] \to \frac{1}{N}(x \circledast_N h)[n]$
real x	$X(-\omega) = X^*(\omega)$; even→real, odd→imaginary
Parseval	$\frac{1}{T}\int_T \lvert f\rvert^2 = \sum \lvert a_k\rvert^2$; $\int \lvert x\rvert^2 = \frac{1}{2\pi}\int \lvert X\rvert^2$; $\frac{1}{N}\sum \lvert x\rvert^2 = \sum \lvert X[k]\rvert^2$

Pairs

$\delta(t) \leftrightarrow 1$, $1 \leftrightarrow 2\pi\delta(\omega)$, $e^{j\omega_0 t} \leftrightarrow 2\pi\delta(\omega-\omega_0)$
$e^{-at}u(t) \leftrightarrow \frac{1}{a+j\omega}$, $a^n u[n] \leftrightarrow \frac{1}{1-ae^{-j\Omega}}$ $(\lvert a\rvert<1)$
pulse $\lvert t\rvert < T_1$: $\frac{2\sin\omega T_1}{\omega}$; triangle $1-\lvert t\rvert$: $\left(\frac{\sin\omega/2}{\omega/2}\right)^2$
ideal LPF $\lvert\omega\rvert < W$: $\frac{\sin Wt}{\pi t}$; DT pulse $0..L-1$: $e^{-j\Omega\frac{L-1}{2}}\frac{\sin(L\Omega/2)}{\sin(\Omega/2)}$
periodic f: $X(\omega) = \sum_k 2\pi a_k\delta(\omega-k\omega_0)$
square duty D centered: $a_k = \frac{\sin\pi kD}{\pi k}$; ±1 square: $\frac{2}{j\pi k}$ odd k

Sums and identities

$\sum_{n=0}^{N-1} z^n = \frac{1-z^N}{1-z}$, $\sum_{n\ge0} z^n = \frac{1}{1-z}$
$\sum_{n=0}^{N-1} e^{-j2\pi mn/N} = N$ if $N \mid m$, else 0
$\cos\theta = \frac{e^{j\theta}+e^{-j\theta}}{2}$, $\sin\theta = \frac{e^{j\theta}-e^{-j\theta}}{2j}$, $1-e^{-j2\theta} = 2je^{-j\theta}\sin\theta$
$\cos^2 = \frac{1}{2}+\frac{1}{2}\cos 2\theta$, $\cos^3 = \frac{3}{4}\cos\theta+\frac{1}{4}\cos 3\theta$

Sampling

$x[n] = x(nT_s)$, $\Omega = \omega T_s = 2\pi f/f_s$; DT freq only mod 2π
Nyquist: $f_s > 2f_{max}$; f aliases to $\lvert f-mf_s\rvert$ in $[0, f_s/2]$
$\cos(2\pi pn/q)$ period $q/\gcd(p,q)$

Systems

additive: $x_1+x_2 \to y_1+y_2$; homogeneous: $cx \to cy$ (try $c=-1, 0$); TI: $x[n-n_0] \to y[n-n_0]$
red flags: constants $(+1)$, x^2, $\lvert x\rvert$, $\sin x$ (nonlinear); explicit n, $x[-n]$, $x[2n]$, $x\cos\omega t$ (time-varying)
LTI: $y = x*h$; cascade h_1*h_2, H_1H_2; parallel h_1+h_2
$\sum a_k y[n-k] = \sum b_k x[n-k] \Rightarrow H(\Omega) = \frac{\sum b_k e^{-jk\Omega}}{\sum a_k e^{-jk\Omega}}$
$y+\tau y' = x \Rightarrow H = \frac{1}{1+j\omega\tau}$

Convolution

$(x*h)[n] = \sum_k x[k]h[n-k]$: add shifted copies of x scaled by $h[k]$
support: starts add, ends add; length L_1+L_2-1; sums (areas) multiply
$u*u = (n+1)u[n]$, $u*u*u = \frac{(n+1)(n+2)}{2}$, $a^n u*u = \frac{1-a^{n+1}}{1-a}$
circular N: linear conv, then wrap mod N (no wrap if $N \ge L_1+L_2-1$)

Frequency response

$e^{j\Omega n} \to H(\Omega)e^{j\Omega n}$; $\cos\Omega_0 n \to \lvert H\rvert\cos(\Omega_0 n+\angle H)$
$= \mathrm{Re}\,H\cos\Omega_0 n - \mathrm{Im}\,H\sin\Omega_0 n$
$H(0) = \sum h$, $H(\pi) = \sum(-1)^n h[n]$; linear angle slope $-d$ = delay d; jumps of π = sign changes

Modulation (AM)

$y = x\cos\omega_c t$: copies at $\pm\omega_c$, band $\omega_c \pm W$; need $\omega_c > W$
demod: $\times\cos\omega_c t$, LPF gain 2, cutoff in $(W, 2\omega_c - W)$; phase error $\theta \Rightarrow \times\cos\theta$

DFT practice

bin k at $f = kf_s/N$; spacing f_s/N; window N/f_s s
real cos on bin m: spikes $\frac{1}{2}$ at $k = m, N-m$; off-bin: leakage
$(-1)^n x[n] \to X[k-\frac{N}{2}]$; zero padding = denser samples, same resolution
main lobe of length-N window: $\pm\frac{2\pi}{N}$